

--Project Story--

I worked on a Blinkit sales analysis project to understand what factors were associated with sales performance across products and outlets.

I started with a raw dataset containing 8,523 records and 12 attributes covering product characteristics, outlet characteristics, sales, ratings, visibility and item weight.

The first challenge was data quality. The `Item Fat Content` column contained inconsistent representations such as `LF`, `low fat`, `Low Fat`, `reg` and `Regular`. I standardized these categories into two consistent groups — Low Fat and Regular — before performing the analysis.

I then established four core KPIs: total sales, average sales, number of unique items and average rating. The dataset generated approximately \$1.2 million in sales, with an average sales value of around \$141 and an average rating of approximately 4.0.

From there, I broke sales down across multiple business dimensions.

The analysis showed that Low Fat products contributed 64.6% of sales, while Regular products contributed 35.4%. At the category level, Fruits & Vegetables and Snack Foods were the strongest contributors.

Geographically, Tier 3 locations generated the highest sales, followed by Tier 2 and Tier 1. I also discovered that Medium-sized outlets contributed the largest share of sales, which was an interesting reminder that larger physical outlets do not automatically produce higher aggregate sales.

I also analyzed outlet establishment years. Rather than simply claiming that older outlets perform better, I noticed that the number of outlets differs across establishment years. This helped me understand an important principle of analytics: correlation in a visualization isn't automatically causation.

The project helped me practice the complete analytical workflow — from data cleaning and exploratory analysis to KPI development, visualization and business interpretation.

More importantly, it changed how I look at data. Instead of asking only *"What does the chart show?"*, I started asking:

"Why does it show this, what could explain it, and what business decision could this insight support?"

That is the approach I want to continue developing as I move further into data analytics and business intelligence.